

Statistics & Probability

ANSWER KEY



Mean, median, and mode

- Find the mean and median of the data set: 12, 15, 18, 20, 25, 30.
Mean = $\frac{12 + 15 + 18 + 20 + 25 + 30}{6} = \frac{120}{6} = 20$. Median (average of the two middle values): $\frac{18 + 20}{2} = 19$.
- Find the mode and range of the data set: 7, 9, 9, 11, 14, 14, 14, 18.
Mode: 14 (appears most frequently, three times). Range = $18 - 7 = 11$.
- Find the mean of the data set: 8, 11, 13, 16, 22.
Mean = $\frac{8 + 11 + 13 + 16 + 22}{5} = \frac{70}{5} = 14$.

Variance and standard deviation

- Find the variance and standard deviation of the data set 4, 6, 8, 10, 12, using $\sigma^2 = \frac{\sum(x - \bar{x})^2}{n}$.
Mean = 8. Deviations squared: 16, 4, 0, 4, 16, sum = 40. Variance = $\frac{40}{5} = 8$. Standard deviation = $\sqrt{8} = 2\sqrt{2} \approx 2.83$.
- Find the variance and standard deviation of the data set 10, 12, 14, 18, 21, using $\sigma^2 = \frac{\sum(x - \bar{x})^2}{n}$.
Mean = $\frac{10 + 12 + 14 + 18 + 21}{5} = \frac{75}{5} = 15$. Deviations squared: 25, 9, 1, 9, 36, sum = 80. Variance = $\frac{80}{5} = 16$. Standard deviation = $\sqrt{16} = 4$.

Basic probability

- A bag contains 5 red, 3 blue, and 2 green counters. Find the probability of drawing a blue counter.
Total counters: 10. $P(\text{blue}) = \frac{3}{10}$.
- Using the same bag (5 red, 3 blue, 2 green counters), find the probability of NOT drawing a red counter.
 $P(\text{red}) = \frac{5}{10} = \frac{1}{2}$, so $P(\text{not red}) = 1 - \frac{1}{2} = \frac{1}{2}$.
- Using the same bag (5 red, 3 blue, 2 green counters), find the probability of drawing either a red or a green counter.
 $P(\text{red or green}) = \frac{5 + 2}{10} = \frac{7}{10}$.

The addition and multiplication rules

- Two events A and B satisfy $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.
 $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7$.

- 10 Two independent events A and B have $P(A) = 0.3$ and $P(B) = 0.6$. Find $P(A \cap B)$.
Since A and B are independent: $P(A \cap B) = P(A) \times P(B) = 0.3 \times 0.6 = 0.18$.
- 11 Two events A and B satisfy $P(A) = 0.35$, $P(B) = 0.45$, and $P(A \cup B) = 0.65$. Find $P(A \cap B)$.
 $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.35 + 0.45 - 0.65 = 0.15$.

Conditional probability from a two-way table

- 12 A survey of 200 students found 90 study French, 70 study Spanish, and 30 study both. Find the probability a student studies Spanish, given that they study French.
 $P(\text{Spanish} \mid \text{French}) = \frac{P(\text{both})}{P(\text{French})} = \frac{30/200}{90/200} = \frac{30}{90} = \frac{1}{3}$.
- 13 In the same survey of 200 students (90 study French, 70 study Spanish, 30 study both), find the probability that a student studies neither language.
Number studying at least one language: $90 + 70 - 30 = 130$. Number studying neither: $200 - 130 = 70$.
 $P(\text{neither}) = \frac{70}{200} = \frac{7}{20} = 0.35$.

Tree diagrams

- 14 A box has 4 red and 6 blue balls. Two balls are drawn without replacement. Find the probability both are red.
 $P(\text{first red}) = \frac{4}{10}$. $P(\text{second red} \mid \text{first red}) = \frac{3}{9}$. $P(\text{both red}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.
- 15 Using the same box (4 red, 6 blue balls, drawn without replacement), find the probability that exactly one of the two balls drawn is red.
 $P(\text{red then blue}) = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$. $P(\text{blue then red}) = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$.
 $P(\text{exactly one red}) = \frac{24}{90} + \frac{24}{90} = \frac{48}{90} = \frac{8}{15}$.

Reading and interpreting a bar chart

- 16 The bar chart shows average monthly rainfall (mm) for a city over four months. Find the mean monthly rainfall, and find the percent decrease from March to April.
Mean = $\frac{80 + 65 + 90 + 70}{4} = \frac{305}{4} = 76.25$ mm. Percent decrease from March (90) to April (70):
 $\frac{90 - 70}{90} \times 100\% \approx 22.2\%$.
- 17 Using the same rainfall data (Jan 80, Feb 65, Mar 90, Apr 70 mm), find what percent of the four-month total fell in January.
Total rainfall over the four months: $80 + 65 + 90 + 70 = 305$ mm. Percent in January:
 $\frac{80}{305} \times 100\% \approx 26.2\%$.

Cumulative frequency and quartiles

- 18** A data set has lower quartile $Q_1 = 12$, median $Q_2 = 18$, and upper quartile $Q_3 = 25$. Find the interquartile range, and explain what it measures.

$IQR = Q_3 - Q_1 = 25 - 12 = 13$. The IQR measures the spread of the middle 50% of the data, and is more resistant to outliers than the full range.

- 19** A data set has lower quartile $Q_1 = 15$ and upper quartile $Q_3 = 34$. An outlier boundary is defined as any value more than $1.5 \times IQR$ beyond Q_3 . Find this upper boundary.

$IQR = 34 - 15 = 19$. $1.5 \times IQR = 1.5 \times 19 = 28.5$. Upper outlier boundary $= 34 + 28.5 = 62.5$.

A well-designed word problem: quality control

- 20** This question has two parts. A factory has two machines. Machine A produces 60% of the items and has a 2% defect rate; Machine B produces 40% of the items and has a 5% defect rate. (a) Find the overall probability that a randomly selected item is defective. (b) Given that an item is defective, find the probability it came from Machine B.

(a)

$$P(\text{defective}) = P(A)P(\text{defective} | A) + P(B)P(\text{defective} | B) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.02 = 0.032$$

. (b) Using the definition of conditional probability:

$$P(B | \text{defective}) = \frac{P(B)P(\text{defective} | B)}{P(\text{defective})} = \frac{0.02}{0.032} = 0.625, \text{ or } 62.5\%.$$